

AI QUIZ GENERATOR

Comprehensive User Operations & AI Architecture Guide (PDF Manual)

DOCUMENTATION OVERVIEW

This official PDF User Manual provides step-by-step instructions for operating the AI Quiz Generator, configuring OpenAI API keys, performing Vision OCR extractions, utilizing line range selectors, configuring bilingual learning, managing question banks, and understanding deep AI prompting & model selection architectures.

1. Executive Overview & Core Features

The **AI Quiz Generator** converts knowledge documents, technical guides, PDF files, and scanned images into intelligent assessment quizzes automatically using state-of-the-art OpenAI LLM models and Vision OCR processing.

Feature Module	Capability & Utility Description
Multi-Format File Ingestion	Parses <code>.txt</code> , <code>.md</code> , <code>.docx</code> , <code>.pdf</code> , <code>.png</code> , <code>.jpg</code> , <code>.jpeg</code> , <code>.webp</code> , <code>.gif</code> with AI Vision OCR text extraction.
Section & Line Range Selector	Select specific line ranges or sections to generate questions without duplicating existing question banks.
Bilingual Language Settings	Set separate Question Language and Target Learning Language (e.g., learn Japanese via Vietnamese questions).
Question Bank Management	Edit question text, options, and reference answers or delete individual questions with CSV import/export.
Deep AI Integration & Grading	Supports custom prompt instructions, model selection (<code>gpt-4.1-nano</code> , <code>gpt-4o</code> , <code>gpt-4.1</code>), and real-time AI scoring.

2. Deep AI Architecture: Models, Prompting & Custom Directives

Understanding how the AI engine constructs prompts and evaluates responses empowers you to generate higher-quality assessment questions tailored to your exact learning goals.

2.1 Choosing the Right OpenAI Model

AI Model Name	Best Use Case & Strengths	Token & Cost Profile
<code>gpt-4.1-nano</code>	Ultra-fast generation, large batch text files, basic vocabulary lists, rapid quiz creation.	Most cost-effective / lowest latency.
<code>gpt-4o</code> (Default)	Balanced high accuracy, Vision OCR document transcription, multi-lingual translation, nuanced reasoning.	Standard production model.
<code>gpt-4.1</code>	Complex coding problems, multi-step math derivations, detailed essay grading, strict technical evaluation.	Highest intelligence & reasoning capability.

2.2 Dynamic System Prompt Engineering (`ai.ts`)

The application dynamically builds its system prompt by composing multiple specialized instruction modules:

- Complexity Directives:** Controls question depth (Brief, Elaborate, Practical Application, Coding, Language Vocabulary, or Custom Prompt).
- Language Directives:** Enforces strict rules for Question Language (e.g. Vietnamese) and Target Learning Language (e.g. Japanese) to guarantee terms are translated accurately.
- JSON Output Schema Enforcement:** Formats output strictly as valid JSON containing `text` , `type` , `options` , `correctOptionIndex` , and `writtenAnswerReference` .

PROMPT CONSTRUCTION PIPELINE

System Prompt = [Base Quiz Creator Role] + [Complexity Directive] + [Bilingual Language Directive] + [Custom Instructions] + [JSON Schema Enforcement]

2.3 Writing Custom Prompt Instructions

You can supply custom prompt instructions in the Generation Options panel to shape AI behavior. Examples include:

- "Focus 70% of questions on Linux permission commands (*chmod, chown, umask*) and 30% on process signals."
- "Formulate questions using JLPT N3 level Japanese grammar patterns with furigana annotations."
- "Create scenario-based software architecture questions requiring trade-off analysis."

3. Initial Setup & API Key Configuration

Configure your OpenAI API Key upon launch. Keys are saved securely in local browser storage and sent directly to OpenAI endpoints without intermediate servers.

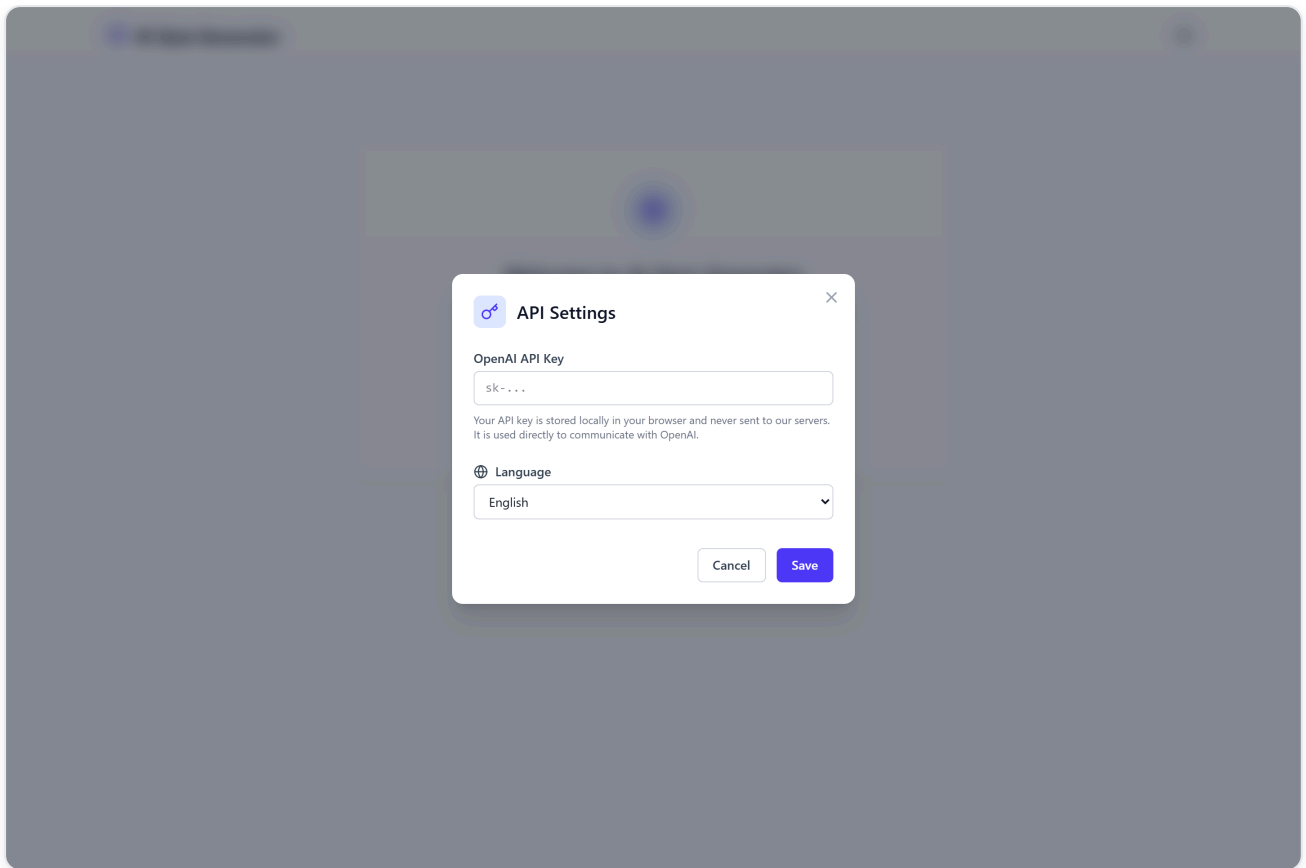


Figure 1: OpenAI API Key Setup Dialog

4. Document Upload & Vision AI OCR Extraction

Drag and drop documents or image scans into the uploader. Vision OCR transcribes handwritten notes and scanned images into editable text before generating questions.

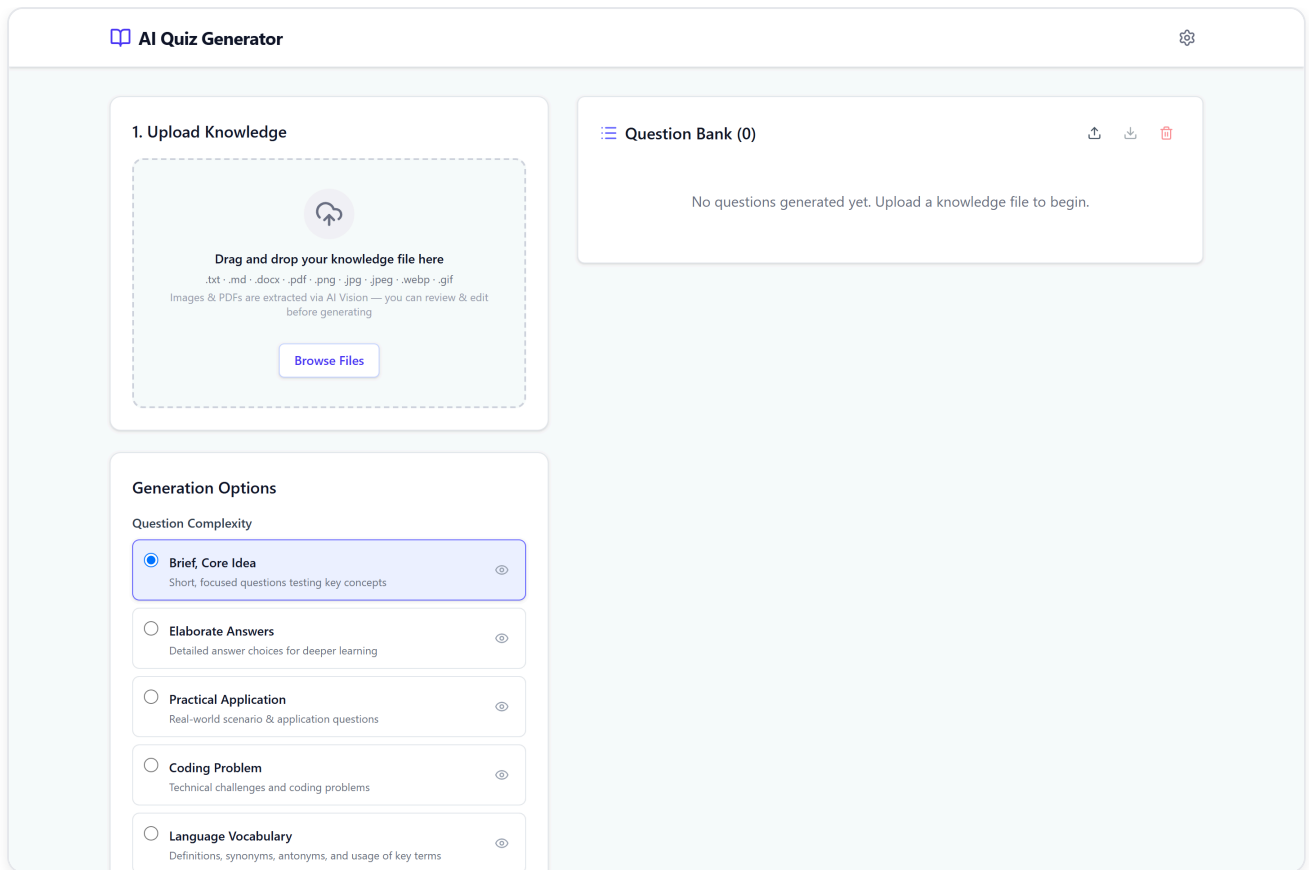


Figure 2: Main Dashboard & Document Upload Zone

5. Section & Line Range Selection (Preventing Question Duplication)

When continuously building a study file, uploading the entire document again can cause question duplication. The **Section & Line Range Selector** allows marking exact line numbers or using presets:

- **Select All Document:** Uses full document text.
- **Bottom Half (Newest):** Targets newly appended content.
- **Last 50 Lines:** Selects only recent additions.
- **Line Click Selection:** Click document lines directly to set ranges.

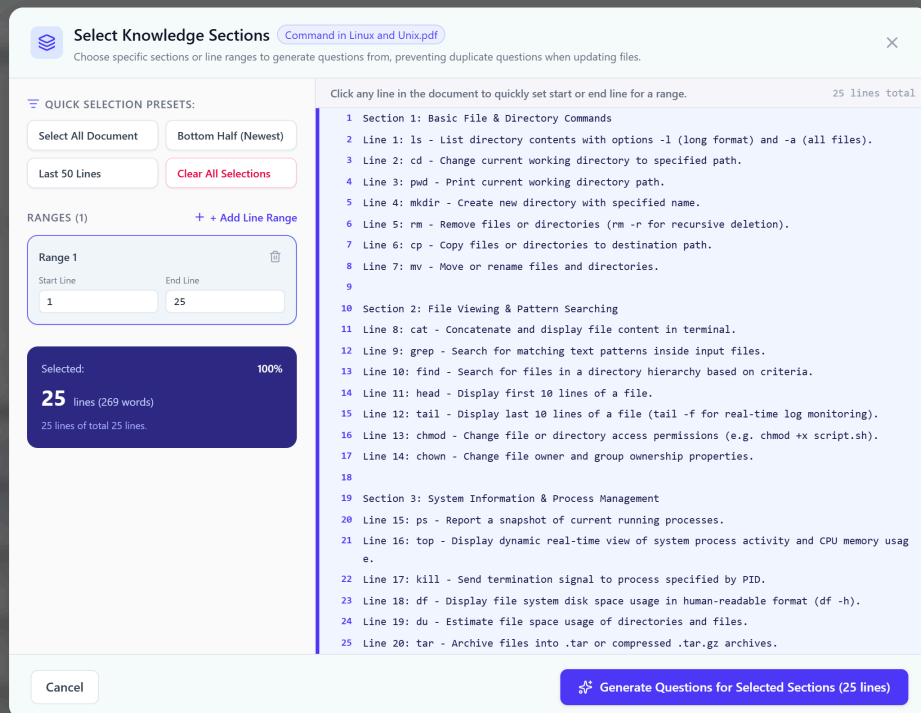


Figure 3: Interactive Knowledge Section & Line Range Selector

6. Generation Options & Bilingual Language Settings

Configure separate Question Language (e.g., Vietnamese) and Target Learning Language (e.g., Japanese) for seamless language learning.

AI Quiz Generator

before generating

Browse Files

Generation Options

Question Complexity

☒ Brief, Core Idea
Short, focused questions testing key concepts

☐ Elaborate Answers
Detailed answer choices for deeper learning

☐ Practical Application
Real-world scenario & application questions

☐ Coding Problem
Technical challenges and coding problems

☐ Language Vocabulary
Definitions, synonyms, antonyms, and usage of key terms

☐ Custom Prompt
Write your own generation instructions with {content} injection

LANGUAGE SETTINGS

Question Language

Auto / Same as document

The language in which questions and explanations are written (e.g. Vietnamese)

Figure 4: Generation Options with Bilingual Language Settings

7. Question Bank Management: Editing & Deleting

All questions are organized in the Question Bank Viewer with individual management buttons:

- **Pencil Icon:** Opens the Edit Question Modal to modify question text, choices, correct options, and written references.
- **Trash Icon:** Permanently deletes individual questions.
- **CSV Import & Export:** Backup question sets or import existing banks.

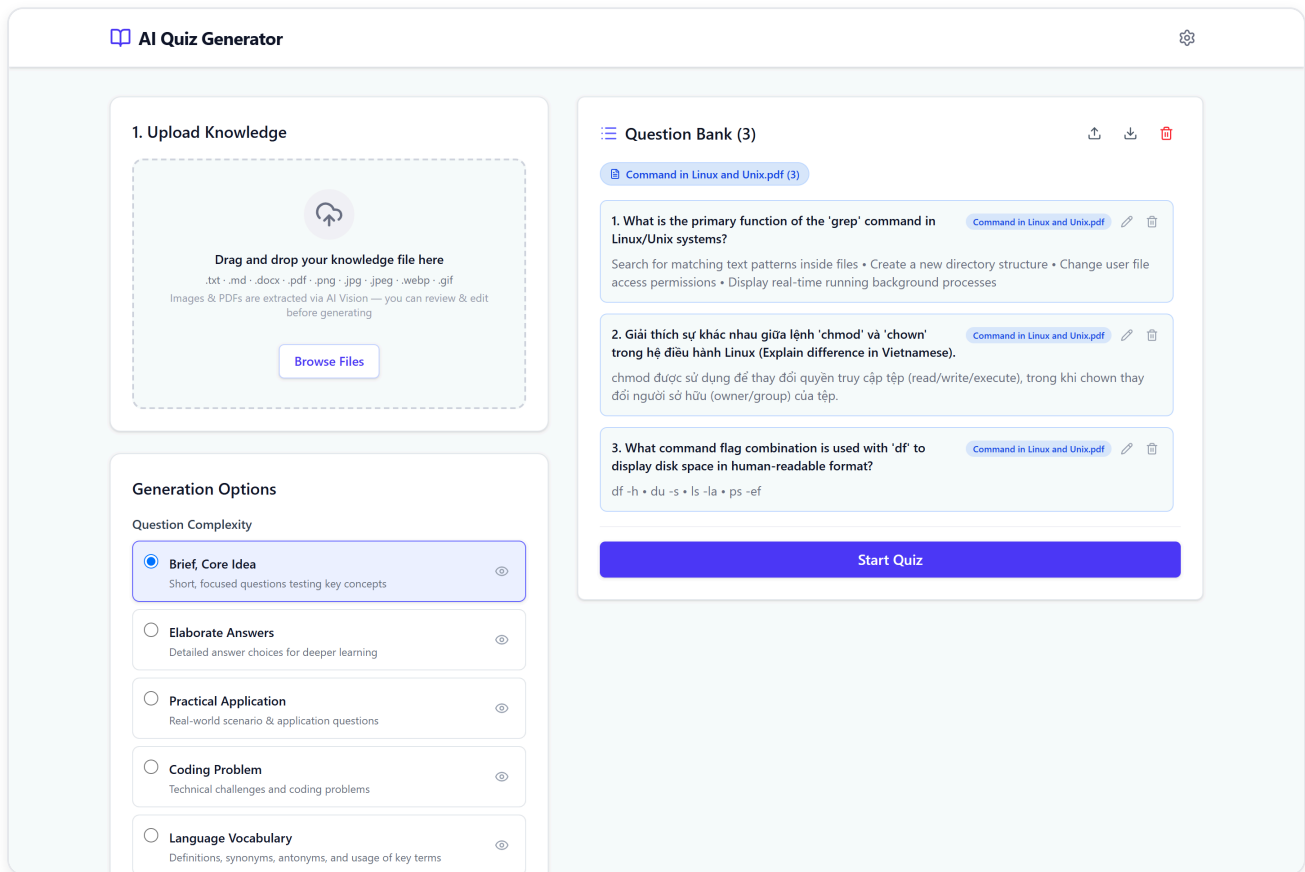


Figure 5: Question Bank Viewer with Edit & Delete Action Buttons

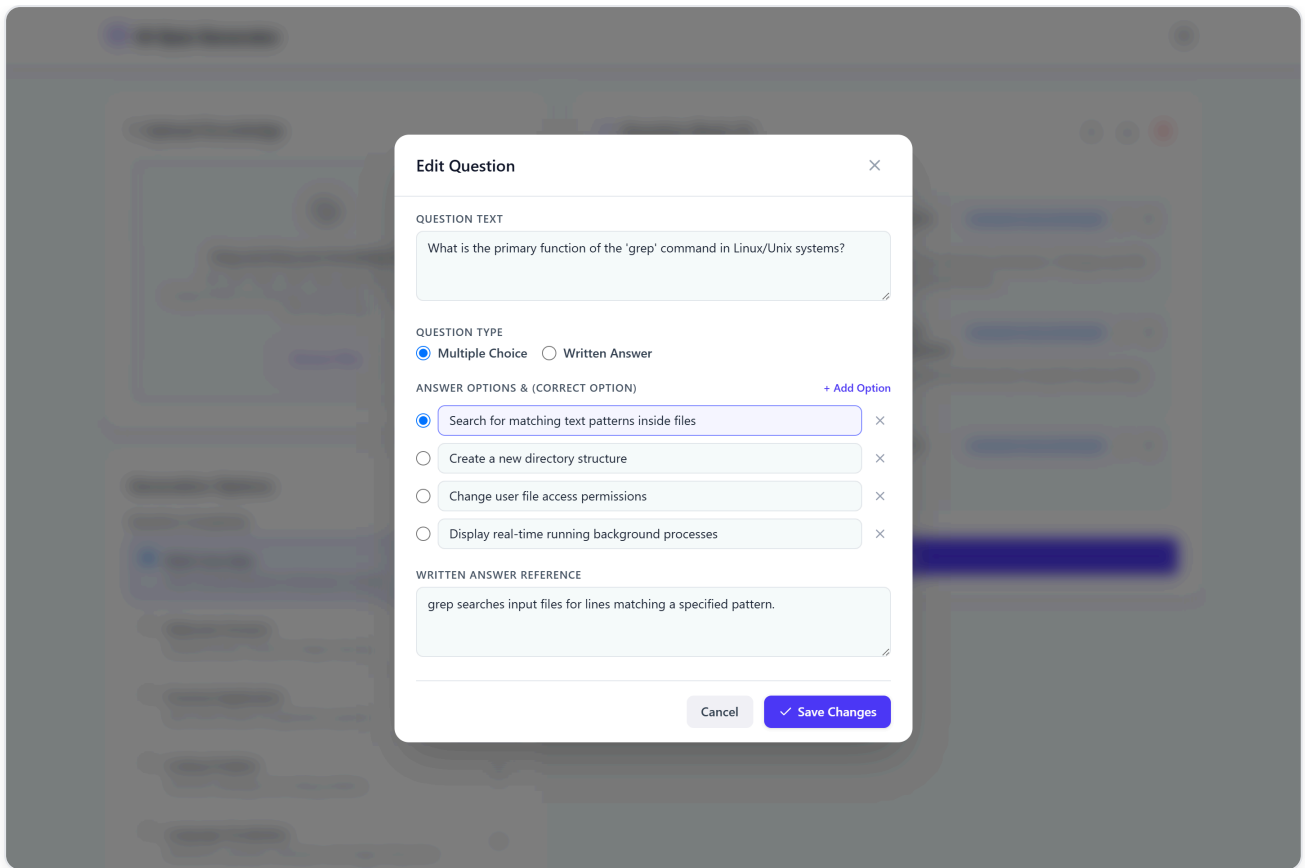


Figure 6: Edit Question Modal Dialog

8. Interactive Quiz Engine & Real-Time AI Evaluation

In written answer or practical coding modes, OpenAI evaluates your answers conceptually, providing constructive score feedback and ideal solutions.

QUESTION

1 of 3

AI EXAMPLE LANGUAGE: JavaScript

Cancel

What is the primary function of the 'grep' command in Linux/Unix systems?

Multiple Choice

Written Answer

<> Code Editor

☐ PRACTICAL EXAMPLE MODE

Search for matching text patterns inside files

Create a new directory structure

Change user file access permissions

Display real-time running background processes

< Previous

Skip >

Submit Answer

Figure 7: Interactive Quiz Interface with Real-Time AI Grading

SUPPORT & FREQUENTLY ASKED QUESTIONS

If you encounter API quota limits or parsing errors, try switching your model selection to `gpt-4.1-nano` or verifying your key credit balance on OpenAI platform.